Notas de charla — IIFET 2026, Tórshavn

Sesión: jueves 20 ago, 13:30–15:00 · Spatial Trade-offs of EBM (Part 2) · hablamos últimos

---

Paper: *Are fishing decisions flexible? Participation, species target, and landing location choices in the U.S. West Coast CPS fishery.* Quezada-Escalona, Tommasi, Kaplan, Muhling & Stohs — *Ecological Economics* **247** (2026).

Sesión: "Spatial Trade-offs of Ecosystem-based Management: Models, Theory, and Performance (Part 2)" — jueves 20 ago, 13:30–15:00, Dansistovan. **Formato real:** 7 papers en 90 min ⇒ ~12 min por slot. Hablamos **últimos**, así que hay que asumir que la sesión va atrasada. **Objetivo:** 10 min de charla + 2 de preguntas. Plan B: 6–7 min (ver §Plan comprimido).

> Cómo usar este documento: la estructura y los comentarios están en español; **las líneas entre comillas son lo que se dice en inglés**, tal cual. No hay que memorizarlas palabra por palabra — sí memorizar la *primera frase* y la *frase de transición* de cada slide, que es donde uno se traba.

Reloj de la charla

#	Slide	Tiempo	Acumulado
1	Título	0:20	0:20
2	Motivation	0:50	1:10
3	**The contribution (hero)**	1:30	2:40
4	Why that swap matters	0:55	3:35
5	Setting	0:40	4:15
6	The model	0:50	5:05
7	Drivers (tabla)	1:00	6:05
8	**Scenario (payoff)**	1:20	7:25
9	Substitution [CUT]	0:45	8:10
10	Takeaways	0:55	9:05
11	Thanks	0:15	9:20

Los dos slides que **no** se sacrifican son el **3** (la contribución) y el **8** (el payoff). Todo lo demás es negociable.

Slide 1 — Título

Idea única: quién soy y de qué va esto, en una frase.

Guion:

> "Thank you. I'm Felipe Quezada, from Universidad de Concepción, and this is joint work with colleagues at NOAA. The paper asks a simple question: **when the ocean changes, how flexible are fishing decisions?** And the way we answer it is by taking a model that

> oceanographers built — a species distribution model — and using its daily output as the central economic regressor.”

Cuidado: no leer la lista de coautores ni los agradecimientos. 20 segundos, se pasa.

Transición: “Let me start with why the *portfolio* of species is the right place to look.”

Slide 2 — Motivation: los trade-offs espaciales pasan por el portafolio de especies

Idea única: el resultado espacial no se decide sólo en el espacio — se decide en *a qué especie te puedes cambiar*.

Guion:

> “Fishable area is shrinking — offshore wind, conservation closures, mining — while the species themselves are moving with climate. For management, what actually matters is not the shock, it’s **how the fleet redistributes effort**: they can follow the species, switch species, switch ports, stay in port, or leave the fishery altogether.
>
> The problem is that location-choice models almost always model **one fishery at a time**. When you do that, the species portfolio drops out of the model by construction — and with it, most of the adaptation margin.
>
> So our question is: how do **availability, prices, costs and weather** shape the **daily** decision to participate, to target a species, and to land at a port? And we ask it in a period with two big natural experiments: the **2015 sardine closure** and the **2014–2016 marine heatwave**.”

Puntero: la columna derecha, las tres palabras en gold (*participate, target, land*).

Transición (memorizar): “So we model those three decisions **jointly** — and the input that makes that possible is on the next slide.”

Slide 3 — La contribución (HERO) ★

Idea única: *el modelo de los oceanógrafos se convierte en el regresor de los economistas.* Éste es **el** slide de la charla. Si sólo se entiende uno, que sea éste. Ir más lento aquí que en cualquier otro.

Guion:

> “Here is what we do differently. Oceanographers build species distribution models — SDMs — for **ecological** purposes: they take the ocean state from a ROMS model, sea surface temperature, sea surface height, chlorophyll, and fit a GAM that predicts the **probability of presence** of a species on a spatial grid, day by day. That’s Muhling and coauthors, 2019 and 2020.
>
> [pausa, seguir la flecha] We take that daily field and average it **within a species-specific radius of each port**. That gives us **Availability**: an **exogenous, gap-free** measure of how much of each species is around **each port, for each species, on each day**. And that variable enters the fisher’s utility function directly.

> One sentence on the assumption: we're assuming **habitat suitability tracks potential catch rates**. If it only does so partly, our availability effect is **attenuated** — biased toward zero, not inflated."

Frase para decir literal, mirando a la audiencia (no a la pantalla):

> "In one line: **the oceanographers' model becomes the economists' regressor.**"

Puntero: recorrer el pipeline izquierda→derecha, arriba→abajo (Ocean state → GAM SDM → daily grid → port radius), y terminar en la caja dorada.

Cuidado:

- El mapa de la derecha **no es output nuestro** — es ilustrativo, de Chasco et al. (2022). Decirlo si alguien pregunta: *"that map is illustrative, it's not our own SDM output — it's there to show what a habitat field looks like."*
- **No** decir que el SDM predice *mejor* la captura que el catch pasado. No hicimos esa comparación (ver §Q&A #1).

Transición: "Why is that swap worth doing? Three reasons."

Slide 4 — Por qué importa el cambio: catch pasado vs. SDM

Idea única: el catch pasado es endógeno, sesgado y con huecos; el SDM no — y sobre todo, el SDM se puede *mover*.

Guion: leer sólo las palabras en negrita, no las viñetas completas.

> "The standard proxy for availability is **past catch** — CPUE from previous trips or previous years. Three problems. It's **endogenous**: it records where vessels already *chose* to fish, and that choice is exactly what we're modeling. It's **selection-biased**: you only observe it at sites that were fished — Chen and co-authors make this point in 2023. And it has **gaps**: it's simply undefined for a port–day when nobody went out.
>
> The SDM index has none of those. It's an **environmental input**, so it's exogenous to the vessel's decision. It has **a value for every port–day**, including the days nobody fished. And — this is the one that buys us the second half of the talk — it's **counterfactual-ready**: you can move the habitat field and re-predict."

Frase de cierre del slide (importante, engancha con el payoff):

> "**You cannot shift a 'past catch' field.** That's the whole reason we can run a spatial scenario at all."

Transición: "Quick word on the setting, and then the results."

Slide 5 — Setting: cuatro segmentos de flota, cuatro portafolios

Idea única: la unidad de adaptación no es "la pesquería", es el **segmento de flota** — cada uno con un portafolio distinto.

Guion: rápido, 40 s. No leer la tabla.

> "This is the U.S. West Coast Coastal Pelagic Species fishery — market squid, sardine, anchovy, mackerels, plus non-CPS options like Dungeness crab, salmon and tuna. Purse seine.
>
> We don't treat it as one fleet. From a cluster analysis in earlier work we get **four segments**: a large southern **squid specialist** — that's the bulk of the data, about 29,000 trips; a **roving squid–sardine generalist**; a **Pacific Northwest sardine specialist**; and a small southern **forage-fish diverse** segment. Four segments, four different portfolios. Same shock, different room to manoeuvre."

Puntero: sólo la columna "Main target(s)".

Números por si preguntan: ~41,000 trips en total (29,160 / 6,806 / 2,581 / 2,481); 2013–2017; los segmentos vienen de Quezada et al. (2024).

Transición: "The model is a standard nested logit — one slide, no derivations."

Slide 6 — El modelo

Idea única: una sola línea de utilidad; lo único que hay que retener es que **Avail entra diario y es exógena**.

Guion: (comprimible a 25 s si vamos tarde)

> "Each alternative is a **triple**: participate, target species, landing port. Utility is linear: an alternative-specific constant, our **availability term** — coefficient allowed to vary by species — expected price, plus distances, wind, closures and local unemployment. And a **state-dependence** term: whether the vessel made that same choice in the last thirty days. That picks up processor ties and port capacity — the fact that these vessels have contracts, not just preferences.
>
> We nest first on **participation**, then on **species** — except for the forage-diverse segment, where the data prefer nesting by **port**. That difference matters later. Crab and salmon are split off, because switching gear there is costly. Estimated with Apollo in R; the dominant species defines the target, and that's 93.6 % of trip revenue, so it's not a coarse assignment."

Cuidado: no derivar nada, no explicar la fórmula del nested logit, no mencionar GEV. Si alguien quiere, está en el backup de λ.

Transición: "So: what actually drives the choice?"

Slide 7 — Drivers: coeficientes de disponibilidad

Idea única: la señal del SDM sobrevive a una inercia muy fuerte ⇒ la flexibilidad que medimos es real.

Guion:

> “Rows are the availability of each species; columns are the four fleet segments. Read **signs and significance**, not magnitudes — these are utility units, so they’re not comparable across segments; magnitude comes from the scenario in two slides.
>
> Squid availability is strongly positive wherever squid is relevant. Anchovy is positive for three segments. Non-CPS availability — crab and salmon — is large and positive for the two segments that have those options: those species **gate participation**.
>
> The row I want you to hold on to is the **mackerels**. Chub and jack mackerel come in **negative in all three southern segments** — minus 3.9 for the squid specialist, minus 1.9 for the forage-diverse fleet, both significant. Mackerel is **not a draw**: it’s what these fleets land when the species they’d rather have isn’t around. Keep that sign in mind — the Los Angeles result two slides from now is made of it.
>
> Now look at the **bottom two rows**, because that’s the whole argument. **Prices** are positive and significant for three of four segments. And **state dependence is big** — between 2.3 and 3.1 everywhere. These fleets are **habitual**: contracts, processors, capacity.”

Frase clave (decirla mirando a la audiencia):

> “And yet, *even with that much inertia in the model*, the **SDM signal survives**. That’s what tells us the flexibility we measure is genuine behaviour, not just persistence.”

Puntero: las **dos filas de abajo**. No pasear el puntero por toda la tabla.

Si preguntan por el anchovy negativo del forage-diverse (−1.88): está en la nota al pie — la disponibilidad de anchoveta y de calamar están **correlacionadas negativamente** (r = −0.29), así que ese coeficiente recoge el costo de oportunidad, no aversión a la anchoveta.

Si preguntan por el mackerel negativo: el mackerel es la **especie de respaldo**, no un objetivo — el signo dice que más hábitat de mackerel **no atrae** esfuerzo. **Ojo con el argumento:** a diferencia de la anchoveta, el hábitat de mackerel está **positivamente** correlacionado con el del calamar en el sur (r ≈ **+0.51** chub, **+0.43** jack, 2013–2017), así que **no** decir “días de mackerel son días sin calamar” — es falso y alguien de la sala lo sabe. Lo defendible: los buenos días de mackerel coinciden con la temporada de calamar, y estas flotas prefieren calamar; el mackerel es de bajo precio y se desembarca cuando no hay algo mejor. **Y este coeficiente es el motor del resultado de LA** (ver slide 8): en la ventana del escenario la disponibilidad de *ambos* mackerels **sube** en Los Angeles (+0.14 chub, +0.16 jack), y con coeficiente negativo eso **baja** la utilidad de esas alternativas. No es sólo sustitución hacia el calamar: es un efecto directo del signo.

Transición (memorizar): “Signs are one thing. Here’s what it’s worth in space.”

Slide 8 — El payoff: mover el campo de hábitat ★

Idea única: movemos el campo de hábitat a una ventana donde el calamar está más al sur, y el modelo re-predice **dónde termina el esfuerzo** — y el total por puerto **esconde una recomposición de especies**.

Guion:

> "This is what the SDM input buys you. We take the **whole daily habitat panel** — every species — and swap it for a **later window within the same season**: two 60-day windows, same months, chosen so that the **squid** centroid sits about **two and a half degrees further south**, 43 North to 40.7. Everything that is *not* habitat is held fixed: prices, costs, distances, weather, state dependence. Then we let the model re-predict.
>
> Each panel is a fleet segment; bars are the change in the **probability of landing at each port**, in percentage points. Gold gains effort, blue loses it.
>
> Two things. First, **Santa Barbara absorbs the squid** — plus 5.8 points for the squid specialist, plus 3.4 for the roving generalist, plus 2.1 for the forage-diverse fleet. And the fleets go out **more often**: non-participation falls by 6 and 4 points for the two southern squid fleets. The Pacific Northwest sardine fleet barely moves — it has no squid to follow.
>
> Second, and this is the interesting one — [apuntar al panel **abajo a la derecha**, *S. CCS forage diverse*] — **Los Angeles goes down**, minus 0.6. But its *squid* landings go **up**, plus 2.0. What it gives up is **mackerel**: chub minus 1.8 and jack minus 0.8, at that same port. And that follows straight from the table you just saw: in this window mackerel habitat improves at Los Angeles as well — but mackerel enters with a **negative** sign, so a better mackerel day is a day these vessels do something else. Mackerel is the species that pays for the squid across the whole south: it's the biggest loser for the squid specialist too, minus 2.2. So the port-level total **hides a recomposition of the species mix.**"

Frase para cerrar (es la que habla directamente a la sesión):

> "And note what this machinery is: we moved a habitat field and got a spatial reallocation. **A closed area is the same operation** — you mask the field inside the polygon and re-predict. That's the door the SDM opens."

Puntero: primero el panel superior izquierdo (squid specialist, la barra grande de Santa Barbara), después el **inferior derecho** (forage diverse) para la historia de LA. **Ojo: la historia de Los Angeles es la del panel forage-diverse**, no la del squid specialist. Aritmética exacta de LA: calamar **+2.02**, chub mackerel **−1.85**, jack mackerel **−0.77**, sardina +0.01 ⇒ **−0.59**. Sin el jack mackerel los números no cierran: hay que nombrar los dos.

Qué se mueve realmente (importante, y está mal contado si se dice "movimos el calamar"): el script intercambia el panel **completo** de SDM diarios — MSQD, PSDN, NANC, JMCK, CMCK, PHRG, ALBC — de la ventana 2016-10-01→11-29 sobre las fechas de la ventana base 2014-10-28→12-26. El calamar es el **criterio** con que se eligen las dos ventanas (centroide más disímil), no lo único que cambia. Δ disponibilidad en Los Angeles: calamar **+0.08**, chub **+0.14**, jack **+0.16**, anchoveta +0.14, sardina +0.08. En el PNW el calamar **no** mejora (CLO −0.005, CWA −0.001): por eso ese segmento casi no se mueve.

Cuidado: el escenario es una **redistribución dentro de la misma temporada**, no una proyección climática. Si se dice "climate scenario", alguien va a preguntar por el forzante y no lo hay. Y **no** decir "movimos el hábitat del calamar" a secas: se mueve el panel entero (ver nota de arriba). Decirlo mal es regalar una pregunta incómoda.

Transición: "Why did LA lose effort while its squid went up? One slide."

Slide 9 — Estructura de sustitución [CUT]

Este es el slide que se bota primero si la sesión va atrasada. *Explica* el anterior, no agrega un resultado nuevo, y el slide de takeaways igual cierra el argumento. Si se bota, decir en su lugar una sola frase al pasar al slide 10: *"and the reason LA behaves that way is that for that fleet, the closest substitute is another species, not another port."*

Idea única: dos formas de adaptarse — *on the move* (seguir la especie entre puertos) y *in place* (cambiar de especie en el mismo puerto).

Guion:

> "The nesting parameters tell you *how* each fleet adapts. Lower lambda means alternatives inside the nest are **closer substitutes**.
>
> For most segments the nests are **species**, and the lambdas are low — anchovy 0.49, sardine 0.40. That means: given the species, the **ports** are close substitutes. Those fleets adapt **on the move** — they follow the species along the coast.
>
> The forage-diverse segment nests by **port**: Santa Barbara 0.49 against 0.73 for Los Angeles and Monterey. There, substitution happens **across species within a port**. That fleet adapts **in place** — and that's why Los Angeles lost effort even as its squid went up: its closest substitute is another species at the *same* port, so when both mackerels lose ground at Los Angeles, the port's own squid can only take part of it back — the rest walks up the coast to Santa Barbara.
>
> The terminology — 'on the move' versus 'in place' — is from Samhouri and co-authors, 2024."

Transición: "Three things to take away."

Slide 10 — Takeaways

Idea única: SDMs sirven como regresores económicos; los pescadores son flexibles; y la respuesta espacial pasa por el portafolio.

Guion:

> "Three points.
>
> **One:** oceanographic SDMs work as **economic regressors** — exogenous, gap-free, daily. And the estimated models predict a **held-out year** well: out-of-sample pseudo-R-squared between **0.22 and 0.44** against the null model.
>
> **Two:** harvesters are **flexible**. Participation, target and landing all respond to availability, prices, costs and weather — and they do so *even after* controlling for very strong state dependence.
>
> **Three:** the spatial answer runs **through the portfolio**. Even the fleet that adapts *in place* — substituting **species** rather than ports — ends up **redistributing effort in space**. If you model one species at a time, you miss that channel entirely.
>
> And because availability here is an **ocean-model output**, the same machinery runs forward under **climate projections**, or under a closed area. Thank you."

✗ **Cuidado crítico:** el 0.22–0.44 es **contra el modelo nulo**, *no* contra una especificación con catch pasado. **Nunca** decir “the SDM predicts better than past catch” — no corrimos esa comparación. El argumento del SDM es de **identificación y de escenarios**, no de superioridad predictiva. (Ver Q&A #1.)

Slide 11 — Thanks

Dejarlo en pantalla durante las preguntas. No leerlo. Sólo: *“Thank you — happy to take questions.”*

Plan comprimido (6–7 min)

Hablamos últimos: es probable que haya que hacer esto. Orden de sacrificio:

1. **Botar el slide 9** (substitution) — pasar directo de 8 a 10 con la frase puente de una línea.
2. **Comprimir el slide 6** (modelo) a 25 s: *“Nested logit over participate × target × port. Availability enters daily and is exogenous; we also control for prices, distances, wind and a thirty-day state-dependence term. Estimated in Apollo.”* y pasar.
3. **Comprimir el slide 5** (setting) a 20 s: *“Four fleet segments from a cluster analysis, four different species portfolios.”*
4. Si aún así falta tiempo: **botar el 4** y meter su idea en una frase dentro del 3 (*“the usual proxy is past catch, which is endogenous, gappy, and — crucially — impossible to shift in a counterfactual”*).

Nunca botar: 3 (contribución) y 8 (payoff). Con esos dos y el 10 la charla se sostiene en 4 minutos.

Números para tener en la cabeza

Dato	Valor
Período	2013–2017 (cierre de sardina 2015; ola de calor 2014–16)
Trips totales	~41,000 (29,160 / 6,806 / 2,581 / 2,481)
Revenue de la especie dominante	93.6 %
State dependence (30 d)	+2.3 a +3.1, *** en los cuatro segmentos
Radios del SDM	60 km (sardina/anchoveta/caballa), 90 km (calamar)
Escenario: centroide del calamar	42.99°N → 40.66°N (mismo período estacional)
Escenario: ventanas	base 2014-10-28→12-26 vs. escenario 2016-10-01→11-29 (60 días, meses 10-11-12)
Escenario: qué se intercambia	**todos** los SDM diarios (MSQD, PSDN, NANC, JMCK, CMCK, PHRG, ALBC); el resto de covariables queda en la ventana base
Δ disponibilidad en LA	calamar +0.08, chub +0.14, jack +0.16, anchoveta +0.14, sardina +0.08
Δ Santa Barbara	+5.8 / +3.4 / — / +2.1 pp

Dato	Valor
Δ Los Angeles (forage diverse)	−0.6 neto = calamar +2.0, chub mackerel −1.8, jack mackerel −0.8
Coef. disponibilidad mackerel	−3.94*** / −1.48 / +0.87 / −1.92*** (negativo en los tres segmentos del sur)
Δ chub mackerel por segmento	−2.2 / −0.3 / +0.1 / −1.9 pp (es la especie que paga el calamar)
Δ participación	+6.0 / +4.2 / −0.6 / +1.9 pp
Δ target calamar	+8.4 (squid spec.) / +4.5 (roving) / +5.0 (forage div.) pp
λ más bajos	anchoveta 0.49, sardina 0.40, Santa Barbara 0.49
OOS pseudo-ρ^2 (año retenido, **vs. nulo**)	0.22–0.44
Correlación anchoveta–calamar (disponibilidad)	r = −0.29 (en los SDM 2013–17: −0.20 a −0.22)
Correlación mackerel–calamar (disponibilidad, sur)	**+0.51** chub, **+0.43** jack — es **positiva**, no negativa

Banco de preguntas

Formato de respuesta: **una frase de respuesta directa, una de sustento, parar.** No dar mini-charlas.

1. "¿Compararon el SDM contra el catch pasado? ¿Predice mejor?" — *La pregunta más probable, y la más fácil de contestar mal.*

> "No — we didn't run that horse race, and I want to be precise about it. Our case for the SDM is **identification and counterfactuals**, not predictive superiority: past catch is endogenous to the choice we're modeling, and you can't shift it in a scenario. What we *do* show is that the estimated models predict a held-out year well against the null, and Table S1 compares SDM **timing** variants — daily, one-day lag, seven-, fourteen- and thirty-day moving averages. A direct SDM-versus-past-catch comparison is a good idea for future work."

2. "Habitat suitability no es lo mismo que tasa de captura."

> "Agreed, and that's our main caveat. Suitability is a proxy for potential catch rates. The good news is the direction of the bias: classical measurement error in a regressor **attenuates** the coefficient. So our availability effects are, if anything, a lower bound."

3. "¿Los precios no son endógenos?"

> "Partly, yes. We use **predicted** prices from a year-trend plus month and port fixed effects, and check robustness with thirty-day moving averages. That reduces simultaneity but doesn't eliminate it, so we read price coefficients as **reduced-form** behavioural responses, not structural supply elasticities."

4. "¿Por qué la anchoveta sale negativa para el forage-diverse?"

> "Because it's an opportunity cost, not a dislike of anchovy. In that region anchovy and squid availability are **negatively correlated** — r of about −0.29 — so high anchovy days are low squid days for a fleet whose best option is squid."

5. “El mackerel sale negativo en casi todas las columnas. ¿Por qué?”

> “Because for these fleets mackerel is the **fallback**, not the target — it’s low-value, often incidental, and the coefficient says a better mackerel field doesn’t pull vessels in. Two things I’d be honest about: mackerel habitat actually *co-moves* with squid habitat in the south, so good mackerel days are peak squid days and the vessels take the squid; and mackerel suitability is probably the weakest of our proxies for catch rates. What the sign does buy us is the scenario result — mackerel habitat improves in the scenario window, and with a negative coefficient that pushes those alternatives down. That’s the Los Angeles story.”

6. “En el escenario, ¿movieron sólo el calamar o todo el campo de hábitat?” — *Es la pregunta natural después del slide 8; contestarla mal deja la impresión de un shock de una sola especie.*

> “The whole habitat panel — every species’ daily field moves together. We pick two 60-day windows in the same months of the year, the pair whose **squid** centroid is most dissimilar, and we transplant the later window’s habitat fields onto the baseline dates day by day. Squid is the **criterion** for choosing the windows, not the only thing that changes. Everything that isn’t habitat — prices, distances, wind, unemployment, state dependence — stays at its observed baseline value. So it’s a coherent ocean state, not a one-species shock; a closed-area counterfactual would be the cleaner single-species experiment, and that’s the natural next run.”

7. “¿Podrían correr un cierre por eólica offshore / un AMP?”

> “Yes — it’s the same operation as the scenario I showed. You mask the habitat field inside the closure polygon, recompute the port-radius averages, and re-predict. We haven’t run it in this paper, but nothing in the machinery has to change. That’s exactly why the exogenous field matters.”

8. “¿Por qué las estructuras de nidos difieren entre segmentos?”

> “It’s empirical. We tried both structures for each segment and kept the one the data supported — fit, and dissimilarity parameters inside the unit interval. For the forage-diverse fleet the port nesting wins, and that’s substantively informative: it *is* the finding about adapting in place.”

9. “¿Agregación a nivel de puerto no oculta heterogeneidad espacial fina?”

> “It does — we list it as our first limitation, following Dépalle and co-authors, 2021. Our choice set is port × species, so within-port location choice is averaged out. The trade-off buys us the joint participation–target–location model; a finer grid would make the choice set intractable.”

10. “¿Cómo separan state dependence de heterogeneidad no observada?”

> “Honestly, not perfectly — that’s the classic initial-conditions problem. What I’d emphasise is the direction: the thirty-day term absorbs a lot of persistence, so it makes it **harder** for availability to show up. The availability effects survive that, which is the point of the slide.”

11. “¿Por qué sólo 2013–2017?”

> “Two reasons: it’s the window where the daily SDM fields and the trip-level PacFIN data overlap cleanly, and it contains the two shocks we care about — the 2015 sardine closure and the 2014–16 marine heatwave.”

12. “¿Modelan entrada y salida de embarcaciones?”

> “No. Our extensive margin is the **daily** decision to go fishing or not, conditional on being in the fleet. Long-run exit and vessel investment are a different model.”

13. “¿Propagan la incertidumbre del SDM?”

> “Not in this paper — availability enters as a fixed regressor. That’s measurement error, so again it attenuates. Propagating the SDM’s predictive distribution through the choice model is a clear next step.”

14. “Los coeficientes de no-CPS son enormes (+10.3 crab en PNW).”

> “They are, and they’re doing a specific job: crab and salmon availability essentially **gate participation** for the fleets that have those permits. They sit in a separate nest because switching gear is costly, so those magnitudes aren’t comparable to the CPS terms.”

15. “¿Y bajo proyecciones climáticas?”

> “The SDMs are driven by ROMS output, so in principle you force them with downscaled projections and run the same behavioural model forward. That’s the direction we’re taking it — with the caveat that the behavioural parameters are assumed stable, which is a strong assumption over decades.”

Backups: mapa rápido

Saber en qué orden están, para llegar de una:

1. **Descriptives** — persistencia + switching (squid specialist: 19.8 cambios de especie, run mediano 2 días). *Úsalo si preguntan por qué hay state dependence.*
2. **Elasticities** — precio mueve la **participación**, disponibilidad mueve el **targeting**. *Úsalo si preguntan por magnitudes (el slide 7 no las da).*
3. **Scenario — participación** (gráfico de barras). *Si preguntan cuánto más salen a pescar.*
4. **Scenario — por especie objetivo**. *Si preguntan qué especie ganó/perdió.*
5. **Scenario — por puerto × especie**. *La mejor para la historia de LA.*
6. **λ (dissimilarity)** — tabla completa. *Si cuestionan la estructura de nidos.*
7. **Variable construction** — SDM, precios, costos, viento. *Si preguntan por los radios o ERA5.*
8. **Limitations** — las cuatro, escritas. *Si la pregunta es incómoda, ir directo aquí: se ve mejor tener la limitación en un slide que improvisarla.*

Recordatorios de entrega

- **Hablamos últimos.** Preguntar al chair al llegar cuánto tiempo real hay. Si dice “eight minutes”, aplicar el plan comprimido **desde el principio**, no improvisar recortes a mitad de charla.
- Mirar a la audiencia en las cuatro frases marcadas (slides 3, 4, 7, 8). El resto se puede leer del apuntador.
- El mapa del slide 3 es **ilustrativo** (Chasco et al. 2022) — decirlo antes de que lo pregunten.
- Nunca afirmar que el SDM **predice mejor** que el catch pasado.
- Si se acaba el tiempo en el slide 8: saltar directo al 10 y decir sólo el punto 3 + la frase de cierre.